

Multi-Asset Trend Following Strategy

Executive Architecture Summary

February 2026 | Complete System Overview

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SYSTEM OVERVIEW

Purpose

A production-grade systematic trend-following strategy that:

- Follows established trends in stocks, ETFs, and crypto
- Uses quantitative rules with zero discretion
- Requires human approval before execution
- Operates on monthly rebalancing cycles
- Manages risk through position sizing and stop-losses

Target Parameters

- Account Size:** €10,000 → €100,000+
- Asset Classes:** US Stocks (NYSE/NASDAQ), EU Stocks (XETRA/LSE/Euronext), Global ETFs, Crypto (BTC/ETH)
- Holding Period:** Medium to long-term (weeks to months)
- Rebalancing:** Monthly (last trading day of month)
- Risk Per Position:** 2% of account equity
- Portfolio Size:** 10-25 positions depending on account size

Core Philosophy

"Follow the trend, cut losses quickly, let winners run."

The strategy identifies instruments in strong uptrends, enters positions with defined risk, and exits when the trend weakens or stops are hit.

BUILDING BLOCKS

1. DATA LAYER (Scripts 1-5)

Script 1: EODHD Bulk Downloader

Purpose: Acquire all market data from EODHD API

Capabilities:

- Downloads OHLCV data for 6 exchanges (NYSE, NASDAQ, XETRA, LSE, PA, AS)
- Three modes: INITIAL (setup), INCREMENTAL (production), CUSTOM (backfill)
- Auto-discovers new symbols (IPOs, new listings)
- Flags delisted symbols
- Downloads corporate actions (splits, dividends)
- Self-healing (automatically catches missed days)

Key Innovation: INCREMENTAL mode auto-detects gap and fills missing data without manual intervention

Script 2: Yahoo Finance Fundamentals Downloader

Purpose: Fetch company metadata

Data Acquired:

- Company name, sector, industry
- Market capitalization
- Currency and exchange
- Employee count

Script 3: Data Consolidator & Validator

Purpose: Merge EODHD + Yahoo data and validate quality

Validation Rules:

- Price consistency (High $\geq$ Low, no 25%+ jumps without corporate actions)
- Completeness (no more than 5% missing bars)
- Volume sanity (no 100 $\times$ spikes without news)
- ATR explosion check (volatility $< 2\times$ median)
- Data freshness (no more than 3 days stale)

Output: Unified time series per symbol with quality flags

Script 4: Universe Screener

Purpose: Apply liquidity and fundamental filters

Filters Applied:

- Minimum market capitalization (varies by region)
- Minimum average daily volume (liquidity requirement)
- Minimum price (avoid penny stocks)
- Minimum listing history (252 trading days)
- Data quality (must pass all validations)
- No pending corporate actions (splits, mergers)

Output: Qualified universe of tradable instruments

Script 5: Indicator Calculator

Purpose: Calculate all technical indicators

Indicators Computed:

- Simple Moving Average 50-day (SMA_50)
- Simple Moving Average 200-day (SMA_200)
- Average True Range 20-day (ATR_20)
- Average Directional Index 14-day (ADX_14)

2. STRATEGY LAYER (Scripts 6-10)

Script 6: Trend Qualifier

Purpose: Identify instruments in confirmed uptrends

Qualification Criteria (ALL must be true):

- $SMA_{50} > SMA_{200}$ (golden cross)
- $Close > SMA_{50}$ (price above short-term trend)
- $ADX_{14} > 20$ (sufficient trend strength)

Output: List of trend-qualified instruments

Script 7: Momentum Ranker

Purpose: Rank qualified trends by momentum strength

Ranking Method:

- Momentum Score = $(Close - SMA_{200}) / SMA_{200} \times 100$
- Sort descending (highest momentum = strongest trend)

Output: Ranked list of instruments by momentum

Script 8: Position Sizer

Purpose: Calculate position sizes adjusted for volatility

Sizing Method:

- Volatility-normalized allocation
- Floor: 0.5% of account
- Ceiling: 8.0% of account
- Target: 4.0% of account (typical)

Output: Share quantities and position values for each instrument

Script 9: Stop-Loss Calculator

Purpose: Calculate initial and trailing stop-loss levels

Stop Types:

- Initial Stop: 3× ATR below entry (set once)
- Trailing Stop: 4× ATR below current price (activates at +15% profit, updates weekly)

Output: Stop prices for all positions

Script 10: Exit Signal Generator

Purpose: Identify positions that should be exited

Exit Triggers (Priority Order):

1. Stop-loss hit (highest priority)
2. Trend reversal ($SMA_{50} < SMA_{200}$)
3. Trend weakness ($ADX < 15$ for 3 consecutive days)
4. Rebalancing rotation (dropped from top N)

Output: List of positions to exit with reasons

3. PORTFOLIO LAYER (Scripts 11-12)

Script 11: Monthly Rebalancer

Purpose: Generate monthly rebalancing recommendations

Process:

1. Check circuit breakers (halt if extreme conditions)
2. Identify mandatory exits (stop-loss, trend reversal)
3. Identify rotation exits (dropped from top N)
4. Select new entries (top N by momentum)
5. Calculate position sizes for new entries
6. Calculate stops for new positions
7. Generate detailed recommendation report

Output: Complete rebalancing plan with exits, entries, and rationale

Script 12: Recommendation Report Generator

Purpose: Create human-readable reports for approval

Report Sections:

- Executive Summary (exits, entries, positions to hold)
- Mandatory Exits (with priority, reason, rationale)
- New Entries (with momentum rank, position size, initial stop)
- Positions to Hold (with current P&L, stop levels)
- Portfolio Allocation (by asset class, by sector)
- Risk Metrics (beta, portfolio ATR, concentration)
- Execution Checklist (day-by-day actions)
- Scenarios (bull/bear/sideways cases)
- Approval Section (sign-off required)

Output: PDF, HTML, and CSV formats

4. EXECUTION & MONITORING LAYER (Scripts 13-14)

Script 13: Execution Logger

Purpose: Record actual trade fills and update portfolio state

Records:

- Trade execution timestamp
- Fill price and quantity
- Execution fees
- Slippage vs. expected price
- Updated portfolio state

Output: Trade log and updated portfolio_state.json

Script 14: Daily Risk Monitor

Purpose: Check portfolio health and trigger alerts

Monitoring Checks:

- Stop-loss violations (any position below stop)
- Portfolio drawdown (alert if > 10%)
- Concentration risk (alert if top 3 positions > 30%)
- Correlation spike (alert if top 3 correlation > 0.85)
- Data staleness (alert if data > 3 days old)

Output: Daily monitoring report with alerts

5. VISUALIZATION LAYER (Script 15)

Script 15: Technical Chart Generator

Purpose: Create interactive charts for analysis

Chart Components:

- Price evolution (200-day history)
- SMA 50 and SMA 200 overlays
- Volume bars
- ADX trend strength indicator
- ATR volatility indicator

Features:

- Interactive navigation (sidebar with search)
- Multiple symbols in single HTML file
- Keyboard shortcuts (arrow keys)
- No external dependencies

Output: Self-contained HTML dashboard

6. BACKTESTING FRAMEWORK (Scripts 16-18)

Script 16: Backtest Engine

Purpose: Test strategy on historical data

Capabilities:

- Walk-forward backtesting (no look-ahead bias)
- Monthly rebalancing simulation
- Transaction cost modeling
- Slippage simulation
- Corporate action adjustments
- Multiple time periods (rolling windows)

Metrics Computed:

- Total return, annualized return, CAGR
- Sharpe ratio, Sortino ratio
- Maximum drawdown, drawdown duration
- Win rate, profit factor
- Average win/loss, win/loss ratio

Output: Backtest results with detailed metrics

Script 17: Walk-Forward Optimizer

Purpose: Optimize parameters without overfitting

Method:

- In-sample period: Train (optimize parameters)
- Out-of-sample period: Test (validate performance)
- Rolling forward through time
- Multiple parameter sets tested

Parameters Optimized:

- SMA periods (50/200 vs 30/150 vs 100/300)
- ADX threshold (15/20/25)
- ATR multipliers (initial: 2.5/3.0/3.5, trailing: 3.5/4.0/4.5)
- Position count (10/15/20/25)

Output: Optimal parameter sets with stability metrics

Script 18: Monte Carlo Simulator

Purpose: Assess strategy robustness to randomness

Simulation Method:

- Randomize trade sequence (preserve individual trades)
- Run 10,000 alternative histories
- Calculate confidence intervals

Metrics Analyzed:

- Distribution of final equity
- Probability of specific drawdown levels
- Worst-case scenarios (5th percentile)
- Best-case scenarios (95th percentile)

Output: Risk distribution and confidence intervals

7. VALIDATION & DECISION LAYER (Scripts 19-21)**Script 19: Backtest Validator**

Purpose: Validate backtest results against acceptance criteria before deployment consideration

10 Primary Validation Tests (All Must Pass):**1. Positive Expectancy Test**

- Threshold: Total Return >30% over 5-year backtest
- Rationale: Must beat cost of capital (3% annually) + inflation (3% annually)
- Pass/Fail: Binary decision

2. Risk-Adjusted Outperformance Test

- Threshold: Strategy Sharpe > Benchmark Sharpe × 1.25
- Benchmark: SPY (S&P 500) buy-and-hold
- Rationale: Must provide superior risk-adjusted returns with buffer for implementation risk

3. Acceptable Drawdown Test

- Threshold: Maximum Drawdown ≤ -30% (absolute limit)
- Target: Maximum Drawdown ≤ -20%
- Excellent: Maximum Drawdown > -15%
- Rationale: Losses beyond -30% cause psychological stress and capital flight

4. Sufficient Trade Count Test

- Threshold: ≥100 completed round-trip trades
- Rationale: Statistical significance requires minimum sample size
- Fail: <100 trades = results could be luck, not skill

5. Realistic Win Rate Test

- Threshold: 35% ≤ Win Rate ≤ 65%
- Suspicious: Win Rate >65% (potential overfitting)
- Fail: Win Rate <35% (too many losses, unsustainable)
- Rationale: 40-50% typical for trend following

6. Positive Profit Factor Test

- Threshold: Profit Factor ≥ 1.5
- Good: Profit Factor ≥ 2.0
- Excellent: Profit Factor ≥ 2.5
- Rationale: Profit factor <1.5 means small edge, vulnerable to cost increases

7. Win/Loss Ratio Test

- Threshold: Average Win ≥ 2.0 × Average Loss

- Good: Ratio ≥ 2.5
- Excellent: Ratio ≥ 3.0
- Rationale: Trend following should have large winners, small losers

8. Transaction Cost Sensitivity Test

- Method: Re-run backtest with 2× transaction costs (0.2% vs 0.1%)
- Threshold: Return drops by <50% when costs double
- Rationale: Strategy must be robust to higher execution costs

9. Drawdown Recovery Time Test

- Threshold: Average recovery ≤ 12 months, Maximum recovery ≤ 24 months
- Good: Average recovery 6-9 months
- Excellent: Average recovery ≤ 6 months
- Rationale: Long recovery times cause investor impatience and capital withdrawal

10. Annual Consistency Test

- Threshold: $\geq 70\%$ of years positive
- Good: 70-80% positive years
- Excellent: $\geq 80\%$ positive years
- Rationale: Strategy shouldn't depend on single outlier year

Performance Scoring System (30 Points Total):

Each metric scored 0-3 points:

- 3 points: Excellent performance
- 2 points: Good/Target performance
- 1 point: Acceptable/Minimum performance
- 0 points: Below minimum (fail)

Metric	Minimum (1 pt)	Target (2 pts)	Excellent (3 pts)
CAGR	8%	12%	18%
Sharpe Ratio	0.8	1.2	2.0
Sortino Ratio	1.0	1.5	2.5
Calmar Ratio	0.5	1.0	2.0
Max Drawdown	-30%	-20%	-15%
Win Rate	35%	45%	55%
Profit Factor	1.5	2.0	2.5
Win/Loss Ratio	2.0	2.5	3.5
Positive Years %	70%	75%	85%
Avg Recovery	12 mo	9 mo	6 mo

Overall Rating Based on Total Score:

- 27-30 points: EXCELLENT (deploy immediately)
- 23-26 points: GOOD (deploy with standard monitoring)
- 19-22 points: ACCEPTABLE (deploy with enhanced monitoring)
- 15-18 points: MARGINAL (paper trade first)
- <15 points: FAIL (do not deploy)

7 Critical Red Flags (Instant Disqualification):

1. Curve-Fitted Equity Curve

- Indicator: >75% positive months, too smooth
- Cause: Look-ahead bias, overfitting
- Action: Reject, review code for bugs

2. Single Trade Dominance

- Indicator: >50% of total return from one trade
- Cause: Luck, not skill; not repeatable
- Action: Reject strategy

3. Excessive Win Rate

- Indicator: Win rate >70% in trend following
- Cause: Overfitting to in-sample data
- Action: Review code, check for data snooping

4. Extreme Parameter Selection

- Indicator: Optimal parameters at grid extremes (edges)
- Cause: Search space too narrow, true optimum outside range
- Action: Expand parameter grid, re-optimize

5. In-Sample vs Out-of-Sample Collapse

- Indicator: Out-of-sample Sharpe <50% of in-sample Sharpe

- Cause: Severe overfitting
- Action: Reject parameters, use simpler model

6. Zero Losing Years

- Indicator: No down years over 5+ year period
- Cause: Unrealistic, cherry-picked data period
- Action: Test on different time periods

7. Unrealistic Trade Count

- Indicator: <100 trades or >2,000 trades over 5 years
- Cause: Parameters too strict or overtrading
- Action: Adjust parameters or question strategy logic

Benchmark Comparisons:

- Primary: SPY (S&P 500) - must beat on risk-adjusted basis
- Secondary: 60/40 Portfolio (60% stocks, 40% bonds)
- Tertiary: Naive Trend Strategy (simple SMA crossover on SPY)

Output:

- Validation report (PDF) with Pass/Fail for each test
- Overall score (0-30 points) with rating
- Red flag analysis with severity assessment
- Benchmark comparison charts
- Recommendation for next steps

Script 20: Out-of-Sample Validator

Purpose: Validate strategy performance on unseen data to detect overfitting

Why Out-of-Sample Validation is Critical:

- In-sample: Parameters optimized on this data (can overfit)
- Out-of-sample: Strategy never saw this data (true test of robustness)
- In-sample results are hypotheses; out-of-sample results are evidence

Walk-Forward Validation Method:

Setup:

- In-Sample Period: 24 months (optimization window)
- Out-of-Sample Period: 6 months (testing window)
- Roll Forward: 6 months (shift window forward)
- Total timeline: 5 years (60 months) = 6 windows

Window Structure:

Window 1: Train on months 1–24, test on months 25–30
Window 2: Train on months 7–30, test on months 31–36
Window 3: Train on months 13–36, test on months 37–42
Window 4: Train on months 19–42, test on months 43–48
Window 5: Train on months 25–48, test on months 49–54
Window 6: Train on months 31–54, test on months 55–60

Process for Each Window:

1. Optimize parameters on in-sample period (find best Sharpe ratio)
2. Lock parameters (no changes allowed)
3. Test on out-of-sample period with locked parameters
4. Record out-of-sample performance
5. Analyze parameter changes between windows

Key Metrics Calculated:

1. Stability Ratio

Stability Ratio = Average OOS Sharpe / Average IS Sharpe

Interpretation:

Excellent: Ratio >0.8 (minimal degradation)

Good: Ratio 0.7–0.8

Acceptable: Ratio 0.6–0.7

Fail: Ratio <0.6 (overfitted)

2. OOS Consistency

Percentage of OOS windows that are profitable

Target: $\geq 70\%$ of windows positive

Good: $\geq 80\%$ of windows positive

Fail: $< 60\%$ of windows positive

3. Parameter Stability

Coefficient of Variation = Std Dev / Mean

For each parameter (e.g., SMA_fast):

Calculate CV across all windows

Excellent: CV $< 10\%$ (very stable)

Good: CV 10–20%

Poor: CV $> 20\%$ (unstable, no clear optimum)

4. Performance Trend Analysis

Plot OOS returns by window

Fit linear regression line

Acceptable: Slope not significantly negative ($p > 0.1$)

Warning: Slope significantly negative (strategy decaying)

Red Flags in Out-of-Sample Validation:

Red Flag 1: Cliff Performance Drop

- In-Sample Sharpe: 2.0
- Out-of-Sample Sharpe: 0.3
- Interpretation: Strategy completely overfitted
- Action: Reject

Red Flag 2: Parameter Instability

- Window 1 optimal SMA: 50
- Window 2 optimal SMA: 200
- Window 3 optimal SMA: 30
- Interpretation: No stable optimal parameters exist
- Action: Simplify strategy, reduce parameters

Red Flag 3: Declining OOS Performance

- 40% of OOS windows are negative
- Interpretation: Strategy not robust to regime changes
- Action: Add regime filters or reject

Holdout Period Validation:

- Reserve last 6-12 months of data (never used in optimization)
- Test on holdout period ONCE after finalizing parameters
- Threshold: Holdout Sharpe $\geq 0.5 \times$ Backtest Sharpe
- This is the ultimate validation test

Output:

- Out-of-sample validation report (PDF)
- Stability ratio for each metric
- OOS consistency percentage
- Parameter convergence charts
- Window-by-window performance table
- Holdout period results
- Pass/Fail recommendation

Script 21: Deployment Decision Engine

Purpose: Make systematic Go/No-Go deployment decision based on all validation results

Monte Carlo Validation Component:

Purpose: Assess whether backtest result was skill or luck

Method:

1. Extract all 500 historical trades from backtest
2. Randomly shuffle trade sequence 10,000 times
3. For each simulation: apply trades in random order, calculate final equity
4. Analyze distribution of 10,000 outcomes

Key Metrics from Monte Carlo:

1. Percentile Rank of Actual Result

Where does actual backtest rank in distribution?

Excellent: 90th+ percentile (very lucky, adjust expectations down)

Good: 70–90th percentile (above average)

Acceptable: 50–70th percentile (average, realistic)

Poor: 30–50th percentile (below average)

Fail: <30th percentile (unlucky or overfitted)

2. 95% Confidence Interval

Range where 95% of outcomes fall

Example: [€48,000, €95,000] from €50,000 starting capital

Interpretation: 95% chance final equity in this range

Narrow range = predictable (good)

Wide range = unpredictable (bad)

Relative Width = (Upper – Lower) / Starting Capital

Excellent: <50%

Good: 50–100%

Poor: >100%

3. Tail Risk Analysis (5th Percentile)

Worst-case scenario (bottom 5% of outcomes)

Acceptable: 5th percentile > –20% loss

Unacceptable: 5th percentile < –30% loss

4. Probability of Target Return

What % of simulations achieved ≥50% return?

Excellent: >75% probability

Good: 60–75% probability

Poor: <50% probability (coin flip)

10-Step Decision Process:

Step 1: Run complete backtest suite

- Script 16: Backtest Engine (5-year historical)
- Script 17: Walk-Forward Optimizer (6 rolling windows)
- Script 18: Monte Carlo Simulator (10,000 runs)

Step 2: Apply 10 primary validation tests (Script 19)

- Record Pass/Fail for each test
- If <8 tests pass → STOP, REJECT strategy immediately

Step 3: Calculate overall score (Script 19)

- Score each metric 0-3 points
- Sum across all 10 metrics (max 30 points)
- Determine rating: EXCELLENT/GOOD/ACCEPTABLE/MARGINAL/FAIL

Step 4: Check for critical red flags (Script 19)

- Review all 7 red flag criteria
- If any red flag present → Investigate thoroughly, likely REJECT

Step 5: Benchmark comparison (Script 19)

- Compare to SPY, 60/40 portfolio, naive trend
- Must beat SPY on risk-adjusted basis (Sharpe ratio)
- Count how many benchmarks beaten

Step 6: Out-of-sample validation (Script 20)

- Check walk-forward results
- Calculate stability ratio
- Verify OOS consistency
- Analyze parameter stability

Step 7: Monte Carlo validation (Script 18 + Script 21)

- Check percentile rank
- Analyze tail risk (5th percentile)
- Verify confidence interval width
- Assess probability of target return

Step 8: Apply 5-Tier Decision Matrix

- Synthesize all evidence from Steps 1-7
- Assign strategy to deployment tier

Step 9: Generate comprehensive validation report

- Executive summary with tier assignment
- Detailed test results
- Comparative analysis vs benchmarks
- Risk assessment and stress tests
- Recommendation with rationale and caveats

Step 10: Obtain required approvals

- Portfolio Manager (strategy logic, expected returns)
- Risk Manager (risk limits, stress test results)
- Compliance (regulatory review)
- All three must approve before deployment

5-Tier Decision Matrix (Detailed):

Tier 1: DEPLOY IMMEDIATELY

Requirements (ALL must be met):

- ✓ All 10 primary tests PASS
- ✓ Overall score ≥ 23 points (GOOD or better)
- ✓ Beats SPY on risk-adjusted basis
- ✓ No critical red flags
- ✓ Stability ratio > 0.7
- ✓ Parameter stability confirmed (CV $< 20\%$)
- ✓ Monte Carlo 5th percentile $> -20\%$

Deployment:

- Full capital allocation to strategy
- Standard position sizes as calculated
- Monthly monitoring frequency
- Quarterly re-validation

Risk Level: LOW
Expected Success Rate: 85–95%

Tier 2: DEPLOY WITH MONITORING

Requirements:

- ✓ All 10 primary tests PASS
- ✓ Overall score 19–22 points (ACCEPTABLE)
- ✓ Beats benchmark on risk-adjusted basis
- ✓ No critical red flags
- ✓ 1–2 warning signs present
- ✓ Stability ratio 0.6–0.7

Deployment Conditions:

- Reduce initial position sizes by 10–20%
- Tighten stop-losses by 10%
- Weekly monitoring (first 3 months)
- Monthly monitoring thereafter
- Quarterly re-validation required
- Conservative ramp-up (increase sizes gradually)

Risk Level: MODERATE
Expected Success Rate: 70–85%

Tier 3: PAPER TRADE FIRST

Requirements:

- ✓ 8-9 out of 10 primary tests PASS
- ✓ Overall score 15-18 points (MARGINAL)
- x Barely beats benchmark OR
- x 2-3 warning signs present OR
- x Stability ratio 0.5-0.6

Deployment:

- Paper trade for 3-6 months
- Generate real-time recommendations but don't execute
- Track how recommendations would perform
- Monitor slippage vs backtest

Success Criteria for Moving to Tier 2:

- Paper trading results match backtest within 20%
- No new red flags emerge
- All warning signs investigated and explained
- Sharpe ratio in paper trading ≥ 0.5

If Successful: Move to Tier 2

If Unsuccessful: Move to Tier 4

Risk Level: MODERATE-HIGH

Expected Success Rate: 50-70%

Tier 4: IMPROVE AND RETEST

Requirements:

- x <8 primary tests pass OR
- x Overall score <15 points OR
- x Does not beat benchmark OR
- x 1+ critical red flags present

Action: DO NOT DEPLOY

Improvement Steps:

1. Identify root cause of failures
2. Simplify strategy (reduce parameters)
3. Expand backtest period (test on more data)
4. Adjust transaction cost assumptions
5. Re-optimize with broader parameter ranges
6. Address specific red flags

Loop: Repeat Steps 1-7 until reaches Tier 3 or higher

Abort: If cannot reach Tier 3 after 3 iterations, abandon strategy

Risk Level: HIGH

Expected Success Rate: 30-50% (after improvements)

Tier 5: REJECT

Requirements:

- x Multiple critical red flags OR
- x Fundamental logic errors OR
- x Cannot be profitable with realistic costs OR
- x Ethical concerns (market manipulation, etc.)

Action: REJECT permanently

Do not attempt to fix. Start over with different approach.

Document reasons for rejection for future reference.

Expected Success Rate: 0% (not fixable)

Required Documentation (5 Reports):

1. Backtest Report (PDF)

- Strategy description, parameters
- All performance metrics
- Equity curve and drawdown charts
- Complete trade list
- Monthly return table

2. Validation Report (PDF)

- All 10 primary test results (Pass/Fail)
- Scoring breakdown (30-point system)
- Red flag analysis with severity

- Benchmark comparison charts
- Final recommendation

3. Parameter Selection Log (CSV + PDF)

- All parameter combinations tested (972 combinations)
- Optimization results for each
- Selected parameters with justification
- Sensitivity analysis ($\pm 20\%$ parameter changes)
- Parameter stability across windows

4. Approval Record (PDF with digital signatures)

- Complete validation report attached
- Sign-off from Portfolio Manager
- Sign-off from Risk Manager
- Sign-off from Compliance
- Date of approval
- Conditions and caveats noted
- Reassessment schedule specified

5. Deployment Checklist (PDF)

- Pre-deployment verification completed
- Configuration confirmed
- Initial position sizes calculated
- Monitoring setup configured
- Alert thresholds set
- Emergency procedures documented

Retention Policy:

- All documents retained indefinitely
- Complete audit trail for regulatory compliance
- Annual audit by external auditor

Output from Script 21:

- Deployment Decision Report (PDF, 30-50 pages)
 - Executive summary (2 pages)
 - Tier assignment with detailed rationale
 - Complete validation results
 - Monte Carlo analysis
 - Risk assessment
 - Deployment conditions
 - Required approvals section
 - Sign-off page
- Decision Dashboard (HTML)
 - Interactive visualization of all metrics
 - Traffic light indicators (red/yellow/green)
 - Drill-down capability into each test
 - Comparison charts vs benchmarks

8. PERFORMANCE MANAGEMENT LAYER (Scripts 22-24)

Script 22: Performance Attribution

Purpose: Decompose returns by source

Attribution Analysis:

- Asset class contribution (stocks vs ETFs vs crypto)
- Sector contribution (which sectors added/subtracted value)
- Position-level contribution (which positions drove returns)
- Alpha vs Beta (skill vs market exposure)
- Active return (strategy vs benchmark)

Output: Attribution report showing return sources

Script 23: Risk Analytics

Purpose: Measure portfolio risk characteristics

Risk Metrics:

- Value at Risk (VaR) - 1-day, 5-day, 20-day
- Conditional Value at Risk (CVaR) - expected loss beyond VaR
- Beta to benchmark (SPY, ACWI)
- Tracking error vs benchmark
- Information ratio (active return / tracking error)
- Concentration metrics (Herfindahl index)
- Correlation matrix (top 10 positions)
- Factor exposures (size, value, momentum, quality)

Output: Comprehensive risk report

Script 24: Performance Dashboard

Purpose: Visualize portfolio performance

Dashboard Components:

- Equity curve (cumulative return over time)
- Monthly return heatmap (calendar view)
- Drawdown chart (underwater plot)
- Rolling Sharpe ratio (12-month window)
- Asset allocation pie chart
- Sector allocation bar chart
- Position-level P&L table
- Top winners and losers
- Trade history timeline

Output: Interactive HTML dashboard

DATA ARCHITECTURE

Data Flow



```
↓
Human Approval (PM, Risk, Compliance)
↓
Deploy to Production OR Reject

PERFORMANCE MANAGEMENT FLOW (Live Trading):
[Script 13: Trade Executions]
↓
[Script 22: Performance Attribution] → Return decomposition
↓
[Script 23: Risk Analytics] → Risk metrics
↓
[Script 24: Performance Dashboard] → Visual reporting
```

Data Storage Structure

```
data_cache/
├── raw_bulk/{exchange}/{date}.parquet
├── fundamentals/company_info.json
├── consolidated/{symbol}.parquet
├── indicators/{symbol}_indicators.parquet
├── qualified/qualified_symbols.json
├── signals/momentum_ranked.json
├── portfolio/position_sizes.json
├── metadata/
│   ├── last_update.json
│   ├── new_symbols.jsonl
│   └── delisted_symbols.jsonl
└── backtest/
    ├── results/{backtest_id}.json
    ├── trades/{backtest_id}_trades.csv
    ├── equity_curve/{backtest_id}_equity.csv
    ├── walk_forward/{backtest_id}_windows.json
    └── monte_carlo/{backtest_id}_simulations.json

data/
├── portfolio_state.json
├── executions/{date}_trades.json
└── performance/
    ├── attribution/{date}_attribution.json
    ├── risk/{date}_risk.json
    └── dashboard/{date}_dashboard.html

reports/
├── rebalancing/{YYYY-MM}_recommendations.pdf
├── daily/{date}_monitoring.json
├── charts/technical_analysis_{timestamp}.html
├── backtest/
│   ├── backtest_report_{id}.pdf
│   ├── validation_report_{id}.pdf
│   ├── parameter_selection_{id}.csv
│   ├── approval_record_{id}.pdf
│   └── deployment_checklist_{id}.pdf
├── performance/monthly_performance_{YYYY-MM}.pdf
└── validation/
    ├── primary_tests_{id}.json
    ├── oos_validation_{id}.json
    └── deployment_decision_{id}.pdf
```

TECHNICAL INDICATORS & FORMULAS

1. Simple Moving Average (SMA)

Formula:

$$SMA(n) = (Close[0] + Close[1] + \dots + Close[n-1]) / n$$

Explanation:

Average of the last n closing prices. Smooths out price action to identify trend direction.

Parameters Used:

- SMA_50: 50-day moving average (short-term trend)
- SMA_200: 200-day moving average (long-term trend)

Interpretation:

- SMA_50 > SMA_200 → Uptrend (golden cross)

- $SMA_{50} < SMA_{200} \rightarrow$ Downtrend (death cross)
- $Price > SMA \rightarrow$ Bullish momentum
- $Price < SMA \rightarrow$ Bearish momentum

Why These Parameters:

- 50 and 200 days are industry-standard periods
- Well-tested in academic literature (Faber, 2007)
- Balance between responsiveness and stability

2. Average True Range (ATR)

Formula:

```
True Range (TR) = max(
    High - Low,
    |High - Previous Close|,
    |Low - Previous Close|
)

ATR(n) = SMA(TR, n)

ATR_Percentage = (ATR / Close) × 100
```

Explanation:

Measures volatility by capturing the largest daily price range, including gaps. Expressed as percentage of price for comparability across instruments.

Parameters Used:

- ATR_20: 20-day average true range

Interpretation:

- High ATR $\rightarrow$ High volatility (wider stops needed)
- Low ATR $\rightarrow$ Low volatility (tighter stops possible)
- ATR_Pct typical ranges: 1-3% (low vol) to 5-10% (high vol)

Why This Parameter:

- 20 days captures recent volatility without being too noisy
- Percentage normalization allows cross-instrument comparison
- Used for position sizing and stop-loss placement

3. Average Directional Index (ADX)

Formula:

```
+DM = High[today] - High[yesterday] (if positive, else 0)
-DM = Low[yesterday] - Low[today] (if positive, else 0)

+DI = 100 × EMA(+DM, n) / ATR(n)
-DI = 100 × EMA(-DM, n) / ATR(n)

DX = 100 × |+DI - -DI| / (+DI + -DI)

ADX = EMA(DX, n)
```

Explanation:

Measures trend strength regardless of direction. Higher ADX = stronger trend. Does not indicate trend direction, only strength.

Parameters Used:

- ADX_14: 14-day average directional index

Interpretation:

- $ADX > 25 \rightarrow$ Strong trend
- $ADX\ 20-25 \rightarrow$ Moderate trend
- $ADX < 20 \rightarrow$ Weak trend or ranging market
- Rising ADX $\rightarrow$ Trend strengthening
- Falling ADX $\rightarrow$ Trend weakening

Why This Parameter:

- 14 days is Wilder's original parameter (creator of ADX)
 - Threshold of 20 filters out weak trends
 - Helps avoid whipsaws in ranging markets
-

STRATEGY RULES & FORMULAS

1. Trend Qualification

Rule: ALL conditions must be TRUE

```
Condition 1: SMA_50 > SMA_200 (golden cross exists)
Condition 2: Close > SMA_50 (price above short-term trend)
Condition 3: ADX_14 > 20 (sufficient trend strength)
```

Explanation:

Condition 1 (Golden Cross):

- Ensures long-term trend is up
- SMA_50 crossing above SMA_200 is classic bullish signal
- Filters out instruments in downtrends

Condition 2 (Price Above SMA_50):

- Confirms current momentum is bullish
- Avoids entering when price is pulling back
- Ensures we're buying strength, not weakness

Condition 3 (ADX Threshold):

- Confirms trend has sufficient strength
- $ADX < 20$ indicates ranging/choppy market
- Reduces false signals and whipsaws

Why This Combination:

- Three independent confirmations reduce false positives
 - Each filter addresses different aspect (long-term, short-term, strength)
 - Academic support (Kaufman, 2013; Pruitt & Hill, 1992)
-

2. Momentum Score

Formula:

```
Momentum Score = ((Close - SMA_200) / SMA_200) × 100
```

Explanation:

Percentage distance of current price above the 200-day moving average. Higher score = stronger momentum.

Example:

- Close = €115
- SMA_200 = €100
- Momentum Score = $((115 - 100) / 100) \times 100 = 15.0\%$

Interpretation:

- Score 0-5% → Weak momentum
- Score 5-15% → Moderate momentum
- Score 15-30% → Strong momentum
- Score > 30% → Very strong momentum (potential overextension)

Why This Formula:

- Simple and transparent
- Captures long-term trend strength
- Comparable across all instruments
- Validated in academic research (Moskowitz et al., 2012)

Alternative Considered:

- Rate of Change (3m, 6m, 12m): Too sensitive to entry timing
- Rejected in favor of SMA_200 distance for stability

3. Position Sizing

Formula:

```
Step 1: Base Risk
Base_Risk = Account_Equity × Target_Risk_Per_Position
where Target_Risk_Per_Position = 2.0%

Step 2: Volatility Adjustment
Volatility_Multiplier = Median_ATR / Instrument_ATR

Step 3: Raw Position Value
Raw_Position_Value = Base_Risk × Volatility_Multiplier

Step 4: Apply Floor and Ceiling
Final_Position_Value = max(
    Account_Equity × 0.5%, (floor)
    min(
        Raw_Position_Value,
        Account_Equity × 8.0% (ceiling)
    )
)

Step 5: Convert to Shares
Shares = floor(Final_Position_Value / Instrument_Price)
Actual_Position_Value = Shares × Instrument_Price
```

Explanation:

Step 1 (Base Risk):

- Target 2% of account equity per position
- Industry standard for trend following
- Example: €50,000 account → €1,000 base risk per position

Step 2 (Volatility Adjustment):

- Inverse ATR weighting: Lower volatility → Larger position
- Normalizes risk across instruments
- Example: Median ATR = 2.0%, Instrument ATR = 2.5% → Multiplier = 0.8

Step 3 (Raw Position Value):

- Base risk adjusted for instrument's volatility
- Example: €1,000 × 0.8 = €800

Step 4 (Floor and Ceiling):

- Floor (0.5%): Prevents dust positions
- Ceiling (8.0%): Prevents concentration risk
- Example: Floor = €250, Ceiling = €4,000

Step 5 (Shares):

- Floor operation prevents fractional shares
- Example: €800 / €100 = 8 shares → €800 actual value

Why This Method:

- Risk parity across positions (equal volatility-adjusted risk)
- Prevents over-allocation to low-volatility instruments
- Prevents under-allocation to high-volatility instruments
- Floor ensures positions are meaningful
- Ceiling prevents excessive concentration

4. Initial Stop-Loss

Formula:

```
Initial_Stop = Entry_Price - (Stop_Multiplier × ATR)

where Stop_Multiplier = 3.0
```

Explanation:

Initial stop is set 3 times the ATR below entry price. This gives the position room to fluctuate while limiting maximum loss.

Example:

- Entry Price = €100
- ATR = €2.50
- Initial Stop = €100 - (3.0 × €2.50) = €92.50
- Stop Distance = 7.5%

Characteristics:

- Set once at entry, never moves down
- Tight stop (3× ATR vs 4× for trailing)
- Exit triggered if close ≤ stop_price on ANY day

Why 3× ATR:

- 2× ATR: Too tight, frequent whipsaws
- 4× ATR: Too loose, large losses
- 3× ATR: Balance between protection and flexibility
- Empirical testing shows optimal tradeoff

5. Trailing Stop-Loss

Activation Rule:

```
Trailing stop activates when:  
Current_Price ≥ Entry_Price × 1.15 (i.e., +15% profit)
```

Calculation Formula:

```
New_Trailing_Stop = Current_Price - (Trailing_Multiplier × ATR)  
  
where Trailing_Multiplier = 4.0  
  
Final_Trailing_Stop = max(Current_Trailing_Stop, New_Trailing_Stop)  
(ratchet mechanism - only moves up)
```

Update Frequency:

```
Updated every Friday market close  
Exit if daily close < trailing_stop
```

Explanation:**Activation (+15% Profit):**

- Trailing stop not active until position is profitable
- 15% threshold ensures significant profit before switching to trailing mode
- Prevents premature activation during initial volatility

4× ATR Distance:

- Wider than initial stop (3× ATR)
- Gives winning trades room to breathe
- Reduces risk of being stopped out during normal pullbacks

Ratchet Mechanism:

- Stop only moves UP, never down
- Locks in profits as price rises
- Creates asymmetric risk/reward (limited loss, unlimited gain)

Weekly Updates:

- Updated Friday close only (not daily)
- Reduces noise and overtrading
- Aligns with weekly planning cycle

Example:

```
Week 1:  
- Entry = €100  
- Initial Stop = €92.50 (3× ATR)  
  
Week 4:  
- Current Price = €120 (+20% profit)
```

- Trailing activates ($>15\%$ profit threshold)
- $ATR = €3.00$
- $Trailing\ Stop = €120 - (4.0 \times €3.00) = €108$
- Locked in profit: $€108 - €100 = €8$ per share

Week 8:

- Current Price = €135
- $ATR = €3.50$
- $New\ Trailing = €135 - (4.0 \times €3.50) = €121$
- $Final\ Trailing = \max(€108, €121) = €121$
- Locked in profit: $€121 - €100 = €21$ per share

Why This Design:

- Protects profits without choking winners
- Weekly updates reduce noise
- Wider stop ($4\times$ vs $3\times$) appropriate for trending positions
- Empirical testing shows optimal performance

6. Entry Timing

Rule:

Entry occurs when ALL conditions are TRUE:

1. Today is monthly rebalancing date (last trading day of month)
2. Instrument passed trend qualification
3. Instrument is in top N by momentum score
4. Instrument is not already in portfolio

Execution:

Order Type: Limit order
Limit Price = $Rebalancing_Date_Close \times 1.005$ (close + 0.5%)
Order Placed: End of rebalancing day
Expected Execution: First trading day of new month
Expiration: Cancel if not filled within 2 trading days

Explanation:

Monthly Rebalancing:

- Reduces transaction costs (vs daily rebalancing)
- Aligns with monthly data reporting cycles
- Prevents overtrading

Top N Selection:

- N varies by account size (10-25 positions)
- Only strongest momentum instruments selected
- Natural diversification through momentum spread

Limit Order (Close + 0.5%):

- Protects against gap-ups
- 0.5% buffer allows for normal market movement
- Cancel if price runs away (preserves discipline)

Why These Rules:

- Clear, unambiguous entry criteria
- No discretion or interpretation needed
- Monthly frequency reduces costs while capturing trends

7. Exit Prioritization

Priority 1 (Highest): Stop-Loss Hit

Rule: $Close \leq Stop_Price$
Action: Exit at next market open (market order)
Rationale: Risk management takes absolute priority

Priority 2: Trend Reversal

Rule: $SMA_{50} < SMA_{200}$ (death cross)
Action: Exit at next market open (market order)
Rationale: Long-term trend has turned bearish

Priority 3: Trend Weakness

Rule: $ADX < 15$ for 3 consecutive days
Action: Exit at next market open (market order)
Rationale: Trend has weakened significantly

Priority 4 (Lowest): Rebalancing Rotation

Rule: Instrument dropped from top N by momentum
Action: Exit at month-end close (market order)
Rationale: Better opportunities available

Explanation:

Why Prioritized:

- Higher priority = more urgent exit
- Stop-loss never waits (immediate risk)
- Trend signals wait until month-end unless critical
- Rotation exits only at rebalancing (lowest urgency)

Why These Exit Rules:

- Stop-loss: Limits losses to predetermined level
- Trend reversal: Exits before significant decline
- Trend weakness: Exits during consolidation (dead money)
- Rotation: Reallocates to stronger opportunities

RISK MANAGEMENT FORMULAS

1. Maximum Position Size

Formula:

$\text{Maximum Position Size} = \text{Account_Equity} \times 8.0\%$

Explanation:

No single position can exceed 8% of account equity, regardless of volatility adjustment.

Rationale:

- Prevents concentration risk
- Typical portfolio: $25 \text{ positions} \times 4\% = 100\%$ (target)
- Maximum: $12.5 \text{ positions} \times 8\% = 100\%$ (worst case)
- Ensures diversification

2. Minimum Position Size

Formula:

$\text{Minimum Position Size} = \text{Account_Equity} \times 0.5\%$

Explanation:

Position must be at least 0.5% of account or not taken.

Rationale:

- Prevents dust positions
- Ensures meaningful impact on portfolio
- Transaction costs proportional to position size
- Minimum: $200 \text{ positions} \times 0.5\% = 100\%$ (theoretical max)

3. Asset Class Limits

Formula:

Asset Class Allocation Ranges:

Stocks: 0% – 50% (target: 42%)
ETFs: 0% – 45% (target: 38%)
Crypto: 0% – 20% (target: 15%)
Cash: ≥ 5% (reserve)

Explanation:

Diversification across asset classes reduces correlation and concentration risk.

Rationale:

- Stocks: Primary return driver
- ETFs: Diversification, lower volatility
- Crypto: High return potential, limited exposure
- Cash: Liquidity reserve for opportunities

4. Sector Concentration Limit

Formula:

Maximum Single Sector Allocation = 30%

Explanation:

No more than 30% of portfolio in any single sector (Technology, Healthcare, Finance, etc.).

Rationale:

- Prevents sector-specific risk
- Example: Tech crash doesn't wipe out portfolio
- Forces diversification across economic sectors

5. Correlation Warning Threshold

Formula:

Correlation Warning = Average pairwise correlation of top 3 positions > 0.70

Explanation:

If the top 3 positions have correlation > 0.70, the system triggers a warning flag.

Rationale:

- High correlation = concentrated risk
- Top 3 positions typically represent 12-24% of portfolio
- Correlation > 0.70 means they move together (not diversified)
- Warning prompts review of position selection

6. Portfolio-Level Circuit Breakers

Circuit Breaker 1: Drawdown Limit

Trigger: $(\text{Current_Equity} / \text{Peak_Equity} - 1) < -15\%$
Action: Halt all new entries for 5 trading days
Rationale: Large drawdown indicates strategy not working in current regime

Circuit Breaker 2: VIX Spike

Trigger: $\text{VIX} > 40$
Action: Halt new entries until $\text{VIX} < 30$ for 3 consecutive days
Rationale: Extreme volatility = unreliable signals

Circuit Breaker 3: Correlation Breakdown

Trigger: Pairwise correlation of top 10 positions > 0.85
Action: Halt new entries, flag for review
Rationale: High correlation = concentrated risk, not diversified

Circuit Breaker 4: Concentration Creep

Trigger: Top 3 positions > 30% of portfolio
Action: Halt new entries, force rebalancing
Rationale: Excessive concentration violates risk limits

Circuit Breaker 5: Data Staleness

Trigger: Any position has data > 3 trading days old
Action: Halt all trading, alert operator
Rationale: Stale data = trading blind

Explanation:

Circuit breakers prevent system from executing into disaster during extreme conditions. Human override possible with logged rationale.

BACKTESTING FRAMEWORK

1. Backtesting Principles

No Look-Ahead Bias:

- Only use information available at time of decision
- Calculate indicators using data up to rebalancing date only
- No peeking at future prices

Walk-Forward Methodology:

- Start with initial capital (e.g., €50,000)
- Apply strategy rules at each rebalancing date
- Execute simulated trades at realistic prices
- Update portfolio state
- Move forward to next rebalancing date
- Repeat for entire backtest period

Realistic Execution:

- Entry: Limit order at close + 0.5% (may not fill)
- Exit: Market order at next open (realistic slippage)
- Transaction costs: 0.1% per trade (conservative)
- Slippage: 0.05% for liquid stocks, 0.1% for ETFs

Corporate Action Handling:

- Adjust for stock splits (price and quantity)
- Reinvest dividends (buy more shares)
- Handle delisted stocks (exit at last available price)

2. Performance Metrics

Return Metrics

Total Return:

$$\text{Total Return} = (\text{Final_Equity} - \text{Initial_Equity}) / \text{Initial_Equity} \times 100$$

Annualized Return:

$$\text{Annualized Return} = (\text{Final_Equity} / \text{Initial_Equity})^{(252 / \text{Trading_Days})} - 1 \times 100$$

Compound Annual Growth Rate (CAGR):

$$\text{CAGR} = (\text{Final_Equity} / \text{Initial_Equity})^{(1 / \text{Years})} - 1 \times 100$$

Risk-Adjusted Return Metrics

Sharpe Ratio:

$$\text{Sharpe Ratio} = (\text{Portfolio_Return} - \text{Risk_Free_Rate}) / \text{Portfolio_Volatility}$$

where:

Portfolio_Return = annualized return

$\text{Risk_Free_Rate} = \text{3-month T-bill rate}$
 $\text{Portfolio_Volatility} = \text{standard deviation of daily returns (annualized)}$

Interpretation:

- Sharpe > 2.0: Excellent
- Sharpe 1.0-2.0: Good
- Sharpe 0.5-1.0: Acceptable
- Sharpe < 0.5: Poor

Sortino Ratio:

$\text{Sortino Ratio} = (\text{Portfolio_Return} - \text{Risk_Free_Rate}) / \text{Downside_Deviation}$

where:

$\text{Downside_Deviation} = \text{standard deviation of negative returns only}$

Interpretation:

- Similar to Sharpe but penalizes downside volatility only
- More relevant for investors concerned with losses

Calmar Ratio:

$\text{Calmar Ratio} = \text{CAGR} / \text{Maximum_Drawdown}$

Interpretation:

- Measures return per unit of drawdown risk
- Calmar > 1.0: Good
- Calmar > 2.0: Excellent

Drawdown Metrics

Maximum Drawdown:

$\text{Drawdown}[t] = (\text{Equity}[t] - \text{Peak_Equity}[0:t]) / \text{Peak_Equity}[0:t]$

$\text{Maximum Drawdown} = \min(\text{Drawdown}[t]) \text{ for all } t$

Interpretation:

- Worst peak-to-trough decline
- Example: -25% max drawdown means portfolio lost 25% from peak

Average Drawdown:

$\text{Average Drawdown} = \text{mean of all drawdown periods}$

Drawdown Duration:

$\text{Maximum time from peak to recovery (in days)}$

Trading Metrics

Win Rate:

$\text{Win Rate} = \text{Number_of_Winning_Trades} / \text{Total_Trades} \times 100$

Profit Factor:

$\text{Profit Factor} = \text{Gross_Profit} / \text{Gross_Loss}$

Interpretation:

- Profit Factor > 2.0: Excellent
- Profit Factor 1.5-2.0: Good
- Profit Factor 1.0-1.5: Acceptable
- Profit Factor < 1.0: Losing system

Average Win / Average Loss:

$\text{Avg Win} = \text{mean}(\text{profit of winning trades})$

$\text{Avg Loss} = \text{mean}(\text{loss of losing trades})$

Win/Loss Ratio = Avg Win / Avg Loss

Expectancy:

Expectancy = (Win_Rate × Avg_Win) − (Loss_Rate × Avg_Loss)

Interpretation:

- Expected profit per trade
- Must be positive for profitable system

3. Walk-Forward Optimization

Purpose:

Optimize parameters on in-sample data, validate on out-of-sample data, to avoid overfitting.

Process:

Step 1: Define Parameter Grid

Parameters to optimize:

- SMA Fast: [30, 50, 100]
- SMA Slow: [150, 200, 300]
- ADX Threshold: [15, 20, 25]
- Initial Stop Multiplier: [2.5, 3.0, 3.5]
- Trailing Stop Multiplier: [3.5, 4.0, 4.5]
- Position Count: [10, 15, 20, 25]

Total combinations: $3 \times 3 \times 3 \times 3 \times 3 \times 4 = 972$ parameter sets

Step 2: Walk-Forward Windows

In-Sample Period: 24 months (optimize on this data)
Out-of-Sample Period: 6 months (test on this data)
Roll Forward: 6 months (shift window forward)

Example:

Window 1:

In-Sample: Jan 2020 – Dec 2021 (optimize)
Out-of-Sample: Jan 2022 – Jun 2022 (test)

Window 2:

In-Sample: Jul 2020 – Jun 2022 (optimize)
Out-of-Sample: Jul 2022 – Dec 2022 (test)

Continue rolling...

Step 3: Optimize In-Sample

For each parameter set:

- Run backtest on in-sample period
- Calculate Sharpe ratio
- Select parameter set with highest in-sample Sharpe

Step 4: Validate Out-of-Sample

Use best parameter set from in-sample
Run backtest on out-of-sample period
Record out-of-sample Sharpe ratio

Step 5: Evaluate Stability

Stability Ratio = Out-of-Sample_Sharpe / In-Sample_Sharpe

Interpretation:

- Ratio > 0.8: Robust (parameters generalize well)
- Ratio 0.5–0.8: Acceptable degradation
- Ratio < 0.5: Overfitting (parameters don't generalize)

Step 6: Select Final Parameters

Aggregate all walk-forward windows
Select parameter set with:

- Highest median out-of-sample Sharpe

- Stability ratio > 0.8
- Consistent performance across all windows

Why This Method:

- Prevents overfitting by testing on unseen data
- Mimics real-world deployment (optimize → deploy → test)
- Rolling windows test robustness across regimes
- Stability ratio identifies parameters that generalize

4. Monte Carlo Simulation

Purpose:

Assess strategy robustness by randomizing trade sequence while preserving individual trade outcomes.

Method:

Step 1: Collect Historical Trades

Extract all trades from backtest:

- Entry date, exit date
- Entry price, exit price
- Profit/loss (in %)
- Duration (days)

Example: 500 historical trades over 5 years

Step 2: Randomize Trade Sequence

```
For simulation i (i = 1 to 10,000):
  Randomly shuffle trade order
  Start with initial capital
  For each trade in shuffled order:
    Apply trade return to current equity
    Update current equity
  Record final equity for simulation i
```

Step 3: Calculate Statistics

From 10,000 simulations:

- Median final equity
- 5th percentile (worst case)
- 95th percentile (best case)
- Standard deviation
- Probability of specific outcomes

Example outcomes:

- Probability (Final Equity > €100,000) = 75%
- Probability (Max Drawdown > 25%) = 15%
- Probability (Negative Return) = 5%

Step 4: Confidence Intervals

95% Confidence Interval for Final Equity:

Lower Bound = 2.5th percentile
Upper Bound = 97.5th percentile

Example:

95% confidence interval: [€75,000, €125,000]
Interpretation: 95% chance final equity is in this range

Why This Method:

- Tests sensitivity to trade sequence
- Quantifies luck vs skill
- Provides realistic range of outcomes
- Identifies extreme scenarios (tail risk)

5. Backtesting Validation & Decision Framework

Purpose of Validation

Backtesting produces metrics, but **metrics alone don't determine if a strategy should be deployed**. Validation answers five critical questions:

- 1. **Is the strategy profitable?** (absolute performance)
- 2. **Is the strategy better than alternatives?** (comparative performance)
- 3. **Is the strategy robust?** (stability across time and parameters)
- 4. **Is the strategy realistic?** (accounts for costs and implementation constraints)
- 5. **Should the strategy be deployed?** (Go/No-Go decision)

Validation Philosophy: "A strategy must perform well consistently, realistically, and for the right reasons."

10 Primary Validation Tests (All Must Pass)

Every backtest must pass these tests before deployment consideration:

Test 1: Positive Expectancy

Requirement: Strategy must beat cost of capital + inflation

Metric: Total Return over backtest period
Threshold: Total Return > 30% for 5-year backtest (6% annually)

✓ PASS: Total Return ≥ 30%
✗ FAIL: Total Return < 30%

Test 2: Risk-Adjusted Outperformance

Requirement: Must beat benchmark on risk-adjusted basis

Metric: Sharpe Ratio
Benchmark: SPY (S&P 500) buy-and-hold
Threshold: Strategy Sharpe > Benchmark Sharpe × 1.25

✓ PASS: Strategy Sharpe ≥ Required threshold
✗ FAIL: Strategy Sharpe < Required threshold

Test 3: Acceptable Maximum Drawdown

Requirement: Worst loss must be tolerable

Metric: Maximum Drawdown
Threshold: Max Drawdown ≤ -30% (absolute limit)

✓ EXCELLENT: Max Drawdown > -15%
✓ GOOD: Max Drawdown -15% to -20%
✓ ACCEPTABLE: Max Drawdown -20% to -25%
✗ WARNING: Max Drawdown -25% to -30%
✗ FAIL: Max Drawdown < -30%

Test 4: Sufficient Trade Count

Requirement: Statistical significance requires minimum sample size

Metric: Total number of completed trades
Threshold: ≥ 100 trades

✓ PASS: Trade count ≥ 100
✗ FAIL: Trade count < 100 (insufficient sample)

Test 5: Realistic Win Rate

Requirement: Win rate must be achievable (not suspiciously high)

Metric: Win Rate (% of profitable trades)
Threshold: 35% ≤ Win Rate ≤ 65%

✗ SUSPICIOUS: Win Rate > 65% (potential overfitting)
✓ PASS: Win Rate 35-65%
✗ FAIL: Win Rate < 35% (too many losses)

Test 6: Positive Profit Factor

Requirement: Gross profits must significantly exceed gross losses

Metric: Profit Factor = Gross Profit / Gross Loss
Threshold: Profit Factor ≥ 1.5

✓ EXCELLENT: Profit Factor ≥ 2.5

- ✓ GOOD: Profit Factor 2.0–2.5
- ✓ ACCEPTABLE: Profit Factor 1.5–2.0
- ✗ FAIL: Profit Factor < 1.5

Test 7: Sensible Win/Loss Ratio

Requirement: Winners larger than losers (trend following characteristic)

Metric: Average Win / Average Loss
Threshold: Ratio ≥ 2.0

- ✓ EXCELLENT: Ratio ≥ 3.0
- ✓ GOOD: Ratio 2.5–3.0
- ✓ ACCEPTABLE: Ratio 2.0–2.5
- ✗ FAIL: Ratio < 2.0

Test 8: Transaction Cost Sensitivity

Requirement: Strategy must remain profitable with higher costs

Test: Re-run backtest with 2× transaction costs (0.2% vs 0.1%)
Threshold: Return drops by < 50% when costs double

- ✓ PASS: Performance degradation < 50%
- ✗ FAIL: Performance degradation ≥ 50%

Test 9: Drawdown Recovery Time

Requirement: Must recover from losses in reasonable time

Metric: Average time from drawdown peak to recovery
Threshold: Average recovery ≤ 12 months

- ✓ EXCELLENT: Avg recovery ≤ 6 months
- ✓ GOOD: Avg recovery 6–9 months
- ✓ ACCEPTABLE: Avg recovery 9–12 months
- ✗ FAIL: Avg recovery > 12 months

Test 10: Annual Consistency

Requirement: Not dependent on single outlier year

Metric: Percentage of positive years
Threshold: ≥ 70% of years positive

- ✓ EXCELLENT: ≥ 80% positive years
- ✓ GOOD: 70–80% positive years
- ✗ FAIL: < 70% positive years

Decision Rule: If <8 tests pass → REJECT strategy immediately

Performance Scoring System (30 Points)

Each metric receives 0-3 points based on performance level:

Metric	Minimum (1 pt)	Target (2 pts)	Excellent (3 pts)
CAGR	8%	12%	18%
Sharpe Ratio	0.8	1.2	2.0
Sortino Ratio	1.0	1.5	2.5
Calmar Ratio	0.5	1.0	2.0
Max Drawdown	-30%	-20%	-15%
Win Rate	35%	45%	55%
Profit Factor	1.5	2.0	2.5
Win/Loss Ratio	2.0	2.5	3.5
Positive Years %	70%	75%	85%
Avg Recovery	12 mo	9 mo	6 mo

Overall Rating:

- 27-30 points: EXCELLENT (deploy immediately)
- 23-26 points: GOOD (deploy with standard monitoring)

- **19-22 points:** ACCEPTABLE (deploy with enhanced monitoring)
- **15-18 points:** MARGINAL (paper trade first)
- **< 15 points:** FAIL (do not deploy)

Critical Red Flags (Instant Disqualification)

Any of these red flags requires immediate rejection or thorough investigation:

Red Flag 1: Curve-Fitted Equity Curve

- Equity curve too smooth, lacks realistic volatility
-  75% positive months (unrealistic consistency)
- **Cause:** Look-ahead bias, overfitting

Red Flag 2: Single Trade Dominance

-  50% of total return from one trade
- **Cause:** Luck, not skill; not repeatable

Red Flag 3: Excessive Win Rate

- Win rate >70% in trend following strategy
- **Cause:** Overfitting to in-sample data

Red Flag 4: Extreme Parameter Selection

- Optimal parameters at edge of tested grid
- **Cause:** Search space too narrow, true optimum outside range

Red Flag 5: Out-of-Sample Collapse

- Out-of-sample performance <50% of in-sample
- **Cause:** Severe overfitting

Red Flag 6: Zero Losing Years

- No down years over 5+ year period
- **Cause:** Unrealistic, cherry-picked data period

Red Flag 7: Unrealistic Trade Count

- <100 trades or >2,000 trades over 5 years
- **Cause:** Parameters too strict or overtrading

Action: Any red flag → Investigate thoroughly, likely REJECT

Out-of-Sample Validation Requirements

In-Sample: Data used for parameter optimization (can overfit)

Out-of-Sample: Data never used in optimization (true test)

Walk-Forward Validation Method:

Rolling Windows:

Window 1: Train on 24 months → Test on 6 months
Window 2: Roll forward 6 months, repeat
Continue for 6+ windows

Aggregate all out-of-sample periods

Calculate Stability Ratio = OOS Sharpe / IS Sharpe

Validation Thresholds:

Stability Ratio Requirements:

- ✓ EXCELLENT: Ratio > 0.8 (20% or less degradation)
- ✓ GOOD: Ratio 0.7–0.8
- ✓ ACCEPTABLE: Ratio 0.6–0.7
- ✗ FAIL: Ratio < 0.6 (overfitted)

OOS Consistency:

- ✓ PASS: ≥70% of OOS windows profitable
- ✗ FAIL: <70% of OOS windows profitable

Monte Carlo Validation Requirements

Percentile Rank of Actual Result:

Where does actual backtest rank in 10,000 simulations?

- ✓ EXCELLENT: 90th+ percentile (lucky sequence, adjust expectations down)
- ✓ GOOD: 70-90th percentile
- ✓ ACCEPTABLE: 50-70th percentile
- ✗ POOR: 30-50th percentile
- ✗ FAIL: <30th percentile (unlucky or overfitted)

Tail Risk Analysis:

5th Percentile (reasonable worst case):

- ✓ ACCEPTABLE: 5th percentile > -20%
- ✗ UNACCEPTABLE: 5th percentile < -30%

Confidence Interval Width:

95% Confidence Interval:

- ✓ EXCELLENT: Relative width < 50% of starting capital
- ✓ GOOD: Relative width 50-100%
- ✗ POOR: Relative width > 100% (highly unpredictable)

5-Tier Decision Matrix

Based on validation results, strategies are assigned to one of five deployment tiers:

Tier 1: DEPLOY IMMEDIATELY

Requirements (ALL must be met):

- ✓ All 10 primary tests PASS
- ✓ Overall score $\geq$ 23 points (GOOD or better)
- ✓ Beats SPY on risk-adjusted basis
- ✓ No critical red flags
- ✓ Stability ratio > 0.7
- ✓ Parameter stability confirmed

Action: Deploy to production

Risk Level: LOW

Monitoring: Monthly

Tier 2: DEPLOY WITH MONITORING

Requirements:

- ✓ All 10 primary tests PASS
- ✓ Overall score 19-22 points (ACCEPTABLE)
- ✓ Beats benchmark
- ✓ No critical red flags
- ✓ 1-2 warning signs present
- ✓ Stability ratio 0.6-0.7

Action: Deploy with enhanced monitoring

Risk Level: MODERATE

Monitoring: Weekly (first 3 months), then monthly

Additional: Reduce initial position sizes by 10-20%

Tier 3: PAPER TRADE FIRST

Requirements:

- ✓ 8-9 out of 10 primary tests PASS
- ✓ Overall score 15-18 points (MARGINAL)
- ✗ Barely beats benchmark OR
- ✗ 2-3 warning signs present OR
- ✗ Stability ratio 0.5-0.6

Action: Paper trade for 3-6 months, then re-evaluate

Risk Level: MODERATE-HIGH

Success Criteria: Paper results match backtest within 20%

Tier 4: IMPROVE AND RETEST

Requirements:

- x <8 primary tests pass OR
- x Overall score < 15 points OR
- x Does not beat benchmark OR
- x 1+ critical red flags

Action: DO NOT DEPLOY. Improve strategy:

1. Simplify (reduce parameters)
2. Expand backtest period
3. Adjust cost assumptions
4. Re-optimize with broader ranges
5. Retest and re-evaluate

Loop until reaches Tier 3 or higher

Abort if cannot reach Tier 3 after 3 iterations

Tier 5: REJECT

Requirements:

- x Multiple critical red flags OR
- x Fundamental logic errors OR
- x Cannot be profitable with realistic costs

Action: REJECT permanently

Do not attempt to fix – start over with different approach

Recommendation Process (10-Step Workflow)

Step 1: Run complete backtest suite (Scripts 16-18)

Step 2: Apply 10 primary validation tests

- Record results: Pass/Fail for each
- If <8 tests pass → STOP, REJECT strategy

Step 3: Calculate overall score (0-30 points)

- Score each metric (0-3 points)
- Sum across all 10 metrics

Step 4: Check for critical red flags

- Review all 7 red flag criteria
- If any red flag present → Investigate, likely REJECT

Step 5: Benchmark comparison

- Compare to SPY, 60/40 portfolio, naive trend
- Must beat primary benchmark (SPY) on risk-adjusted basis

Step 6: Out-of-sample validation

- Check walk-forward results
- Calculate stability ratio
- Verify OOS consistency

Step 7: Monte Carlo validation

- Check percentile rank
- Analyze tail risk (5th percentile)
- Verify confidence interval width

Step 8: Make final recommendation

- Synthesize all evidence
- Assign to decision tier (1-5)
- Document rationale and caveats

Step 9: Create validation report (PDF)

- Executive summary with recommendation
- Detailed test results
- Comparative analysis
- Risk assessment
- Approval section

Step 10: Obtain required approvals

- Portfolio Manager (strategy and returns)
- Risk Manager (risk limits and stress tests)
- Compliance (regulatory review)

Required: All three approvals before deployment

Mandatory Documentation

5 Required Reports:

1. **Backtest Report** (PDF)
 - Strategy description, parameters
 - All performance metrics
 - Equity curve and drawdown charts
 - Complete trade list
2. **Validation Report** (PDF)
 - All 10 primary test results
 - Scoring breakdown
 - Red flag analysis
 - Benchmark comparisons
 - Final recommendation
3. **Parameter Selection Log** (CSV + PDF)
 - All parameter combinations tested
 - Optimization results
 - Selected parameters with justification
 - Sensitivity analysis
4. **Approval Record** (PDF with digital signatures)
 - Validation report
 - Sign-offs from PM, Risk Manager, Compliance
 - Date of approval
 - Conditions and caveats
5. **Deployment Checklist** (PDF)
 - Pre-deployment verification
 - Configuration confirmation
 - Initial position sizes
 - Monitoring setup

Retention: All documents retained indefinitely for audit trail

Validation Example

Sample Backtest Results:

CAGR: 14%
Sharpe: 1.3
Max Drawdown: -22%
Win Rate: 46%
Profit Factor: 2.1
Total Trades: 523
Positive Years: 4/5 (80%)

Validation Process:

Primary Tests:

✓ Test 1: Total return 85% > 30% → PASS
✓ Test 2: Sharpe 1.3 > 1.0 → PASS
✓ Test 3: Max DD -22% > -30% → PASS
✓ Test 4: 523 trades > 100 → PASS
✓ Test 5: Win rate 46% in range → PASS
✓ Test 6: Profit factor 2.1 > 1.5 → PASS
✓ Test 7: Win/Loss 2.6 > 2.0 → PASS
✓ Test 8: 2x cost test degradation 35% < 50% → PASS
✓ Test 9: Avg recovery 9mo < 12mo → PASS
✓ Test 10: 80% positive years > 70% → PASS

Result: 10/10 tests passed ✓

Scoring:

CAGR 14%: 2 pts (Good)
Sharpe 1.3: 2 pts (Good)
Max DD -22%: 1 pt (Acceptable)
Win Rate 46%: 2 pts (Good)
...
Total: 19 points → ACCEPTABLE

Red Flags:

- ✓ No curve-fitting
- ✓ No single trade dominance
- ✓ Realistic win rate
- ✓ Parameters stable
- ✓ OOS stability ratio 0.74
- ✓ All checks passed

Result: 0 red flags ✓

Final Decision:

Tier Assignment: TIER 2 (Deploy with Monitoring)

Rationale:

- All validation tests passed
- Score in acceptable range
- No red flags present
- Beats benchmark
- OOS validated

Conditions:

- Weekly monitoring (first 3 months)
- Reduce initial sizes by 10%
- Quarterly re-validation required

PORTFOLIO PERFORMANCE MANAGEMENT

1. Performance Attribution

Purpose:

Decompose portfolio returns to understand what drove performance.

Asset Class Attribution

Formula:

$$\text{Asset_Class_Contribution}[i] = \text{Weight}[i] \times \text{Return}[i]$$

where:

i = asset class (stocks, ETFs, crypto)

$\text{Weight}[i]$ = average weight of asset class over period

$\text{Return}[i]$ = return of asset class over period

$$\text{Total Portfolio Return} = \sum \text{Asset_Class_Contribution}[i]$$

Example:

Stocks: 45% weight × 12% return = +5.4% contribution

ETFs: 40% weight × 8% return = +3.2% contribution

Crypto: 10% weight × 25% return = +2.5% contribution

Cash: 5% weight × 0% return = 0.0% contribution

Total Portfolio Return = 11.1%

Interpretation:

- Identifies which asset classes added/subtracted value
- Helps decide allocation adjustments

Sector Attribution

Formula:

$$\text{Sector_Contribution}[i] = \text{Weight}[i] \times \text{Return}[i]$$

where:
 i = sector (Technology, Healthcare, Financials, etc.)

Example:

Technology: $25\% \times 15\% = +3.75\%$ contribution
Healthcare: $15\% \times 10\% = +1.50\%$ contribution
Financials: $12\% \times 8\% = +0.96\%$ contribution
...

Interpretation:

- Identifies which sectors drove returns
- Highlights sector concentration risks
- Informs sector rotation decisions

Position-Level Attribution

Formula:

$$\text{Position_Contribution} = (\text{Position_Value_End} - \text{Position_Value_Start}) / \text{Portfolio_Value_Start} \times 100$$

Top 10 Contributors: Largest positive contributions
Top 10 Detractors: Largest negative contributions

Example:

Top Contributors:
1. AAPL.US: +2.5% (strong momentum, held full period)
2. NVDA.US: +2.1% (AI rally, trailing stop protected profit)
3. MSFT.US: +1.8% (steady uptrend, full position)

Top Detractors:
1. XYZ.US: -0.8% (stop-loss hit early)
2. ABC.US: -0.5% (trend reversal, timely exit)
3. DEF.US: -0.3% (rotation exit, better opportunity found)

Interpretation:

- Identifies best and worst performers
- Validates entry/exit decisions
- Highlights winners to potentially scale up

2. Alpha vs Beta Decomposition

Purpose:

Separate skill-based returns (alpha) from market exposure returns (beta).

Formula:

$$\text{Portfolio Return} = \text{Alpha} + (\text{Beta} \times \text{Benchmark Return})$$

where:
$$\text{Beta} = \text{Covariance}(\text{Portfolio_Returns}, \text{Benchmark_Returns}) / \text{Variance}(\text{Benchmark_Returns})$$

$$\text{Alpha} = \text{Portfolio Return} - (\text{Beta} \times \text{Benchmark Return})$$

Benchmark Selection:

- US Stocks: S&P 500 (SPY)
- Global Stocks: MSCI All-Country World Index (ACWI)
- Mixed Portfolio: 60% ACWI + 40% Aggregate Bond Index

Example:

Portfolio Return: +12.5%
Benchmark Return (ACWI): +10.0%
Beta: 0.85

$$\text{Beta Contribution} = 0.85 \times 10.0\% = +8.5\%$$

$$\text{Alpha} = 12.5\% - 8.5\% = +4.0\%$$

Interpretation:

- Beta: Market exposure return (passive)
- Alpha: Skill-based return (active management)

- Positive alpha = outperformance vs benchmark
- Target: Alpha > 3% annually

3. Risk Metrics

Value at Risk (VaR)

Formula:

$$\text{VaR} = \text{Portfolio_Value} \times \text{Percentile}(\text{Daily_Returns}, \text{confidence_level})$$

Common confidence levels:

- 95% VaR: Expected loss not exceeded 95% of the time
- 99% VaR: Expected loss not exceeded 99% of the time

Time Horizons:

1-Day VaR: Risk over next trading day
5-Day VaR: Risk over next week
20-Day VaR: Risk over next month

Example:

Portfolio Value: €50,000
Daily Returns: Historical distribution over last 252 days
95% VaR (1-day) = €50,000 × 2.5% = €1,250

Interpretation: 95% confidence that daily loss will not exceed €1,250

Conditional Value at Risk (CVaR)

Formula:

$$\text{CVaR} = \text{Expected loss beyond VaR threshold}$$
$$\text{CVaR} = E[\text{Loss} \mid \text{Loss} > \text{VaR}]$$

Example:

95% VaR = €1,250
CVaR = €1,850

Interpretation: If loss exceeds VaR threshold, expected loss is €1,850

Why CVaR Matters:

- VaR only tells you the threshold
- CVaR tells you expected loss in tail events
- More informative for risk management

Tracking Error

Formula:

$$\text{Tracking Error} = \text{Standard_Deviation}(\text{Portfolio_Returns} - \text{Benchmark_Returns})$$
$$\text{Annualized Tracking Error} = \text{Daily_Tracking_Error} \times \sqrt{252}$$

Example:

Daily tracking error: 0.5%
Annualized tracking error: 0.5% × sqrt(252) = 7.9%

Interpretation: Portfolio deviates from benchmark by ±7.9% annually

Target Ranges:

Tracking Error < 5%: Low active risk (closet indexer)
Tracking Error 5–10%: Moderate active risk
Tracking Error > 10%: High active risk (concentrated bets)

Information Ratio

Formula:

Information Ratio = (Portfolio_Return - Benchmark_Return) / Tracking_Error

= Active_Return / Tracking_Error

Example:

Active Return: +4.0% (alpha)

Tracking Error: 7.9%

Information Ratio: 4.0% / 7.9% = 0.51

Interpretation: 0.51 units of active return per unit of tracking error

Target:

IR > 0.50: Good active management

IR > 0.75: Excellent active management

IR > 1.00: Outstanding (rare)

Concentration Metrics

Herfindahl Index:

HHI = Σ (Weight[i])^2 for all positions

where Weight[i] = position value / portfolio value

Example:

20 positions, equal weight (5% each):

HHI = 20 × (0.05)^2 = 0.05

10 positions, top position 15%, rest 5-8%:

HHI = (0.15)^2 + (0.12)^2 + ... = 0.08

Interpretation:

HHI = 0.04 (25 positions, equal weight): Diversified

HHI = 0.05-0.08: Moderate concentration

HHI > 0.10: High concentration risk

4. Performance Dashboard Components

Equity Curve

Description:
Line chart showing cumulative portfolio value over time.

Formula:

Equity[t] = Equity[t-1] × (1 + Return[t])

Starting with initial capital

Visual Elements:

- X-axis: Time (monthly)
- Y-axis: Portfolio value (€)
- Benchmark overlay (comparison line)
- Drawdown periods shaded

Monthly Return Heatmap

Description:
Calendar view showing returns by month and year.

Layout:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2020	+2.5%	+1.8%	-3.2%	+4.1%	+2.7%	+1.9%	+3.5%	-0.8%	+2.1%	+3.8%	+4.2%	+1.5%
2021	+3.1%	+2.9%	+1.5%	+2.8%	-1.2%	+3.7%	+2.4%	+1.9%	-2.1%	+3.5%	+2.8%	+1.8%
2022	-2.5%	-1.8%	+0.5%	-3.1%	+1.2%	-2.8%	+2.1%	+1.5%	-1.9%	+2.8%	+3.2%	+1.1%

Color Coding:

- Green: Positive returns (darker = higher)
- Red: Negative returns (darker = larger loss)
- Gray: No data

Drawdown Chart (Underwater Plot)

Description:

Area chart showing portfolio drawdown from peak over time.

Formula:

$$\text{Drawdown}[t] = (\text{Equity}[t] - \text{Peak_Equity}[0:t]) / \text{Peak_Equity}[0:t] \times 100$$

Visual Elements:

- X-axis: Time
- Y-axis: Drawdown (%)
- Fill area below zero line
- Marks recovery points (return to 0%)

Rolling Sharpe Ratio

Description:

Line chart showing 12-month rolling Sharpe ratio over time.

Formula:

For each month t:

Calculate returns for months [t-11, t]

$$\text{Sharpe}[t] = (\text{mean}(\text{returns}) - \text{risk_free_rate}) / \text{std}(\text{returns}) \times \text{sqrt}(12)$$

Visual Elements:

- X-axis: Time
- Y-axis: Sharpe ratio
- Threshold line at Sharpe = 1.0 (acceptable)
- Threshold line at Sharpe = 2.0 (excellent)

Asset Allocation Pie Chart

Description:

Current portfolio allocation by asset class.

Segments:

- Stocks (42%)
- ETFs (38%)
- Crypto (15%)
- Cash (5%)

Color Coding:

- Blue: Stocks
- Green: ETFs
- Orange: Crypto
- Gray: Cash

Position-Level P&L Table

Description:

Sortable table showing all current positions.

Columns:

- Symbol
- Entry Date
- Entry Price
- Current Price
- Shares
- Position Value
- Unrealized P&L (€)
- Unrealized P&L (%)
- Current Stop
- Distance to Stop (%)

- Momentum Rank

Sorting:

- Default: By unrealized P&L (%) descending
 - User can sort by any column
-

OPERATIONAL WORKFLOW

Daily Workflow

6:00 AM - Data Update

Automated:

- Run Script 1 (INCREMENTAL mode)
 - Downloads yesterday's data (6 API calls, 30 seconds)
 - Auto-discovers new symbols
 - Flags delisted symbols

6:05 AM - Risk Monitoring

Automated:

- Run Script 14 (Daily Risk Monitor)
 - Check stop-loss violations
 - Check portfolio drawdown
 - Check concentration risk
 - Send alerts if thresholds breached

If Alerts:

Manual:

- Review alerts
- Check market conditions
- Execute emergency exits if needed (stop-loss hits)
- Document decisions

6:30 AM - Performance Review (5 minutes)

Manual:

- Review Script 24 dashboard
 - Check yesterday's P&L
 - Review position performance
 - Check trending positions approaching stops

Weekly Workflow

Friday Close - Stop-Loss Updates

Automated:

- Run Script 9 (Calculate Stops)
 - Update trailing stops (for positions with +15% profit)
 - Stops only move up (ratchet mechanism)
 - Log new stop levels

Friday Evening - Weekly Review (15 minutes)

Manual:

- Review Script 24 dashboard
 - Week's performance summary
 - Check rolling Sharpe ratio
 - Review new/delisted symbols log
 - Identify trending positions vs weakening positions

Monthly Workflow

Last Trading Day of Month - Rebalancing

5:00 AM - Pre-Rebalancing Data Prep

Automated:

- Run Script 1 (INCREMENTAL mode)

- Ensure all data is current
- Run Scripts 2-5 (if new symbols detected)
- Download fundamentals
- Consolidate and validate
- Screen universe
- Calculate indicators

9:00 AM - Generate Rebalancing Recommendations

Automated:

- Run Scripts 6-11
 - Qualify trends
 - Rank by momentum
 - Calculate position sizes
 - Calculate stops
 - Generate exit signals
 - Create rebalancing plan
- Run Script 12
 - Generate recommendation report (PDF + HTML + CSV)

10:00 AM - Human Review (30-60 minutes)

Manual:

- Review recommendation report
 - Check mandatory exits (stop-loss, trend reversal)
 - Review new entries (momentum rank, position size)
 - Validate position sizes (volatility-adjusted)
 - Check portfolio allocation (asset class, sector)
 - Review risk metrics (concentration, correlation)
 - Evaluate scenarios (bull/bear/sideways)
 - Check circuit breakers (not triggered)

Decision: APPROVE / REJECT / MODIFY

If APPROVE:

- Sign report
- Proceed to execution

If REJECT:

- Document reason
- No trades executed

If MODIFY:

- Adjust recommendations
- Re-run Script 11 with adjustments
- Re-review

First Trading Day of New Month - Execution

9:30 AM Market Open - Execute Exits

Manual:

- Execute all exit orders
 - Mandatory exits: Market orders at open
 - Rotation exits: Market orders at open
- Log fills in Script 13

3:30 PM Market Close - Execute Entries

Manual:

- Place entry limit orders
 - Limit price = Close + 0.5%
 - Good for 2 trading days
- Monitor fills over next 2 days
- Log fills in Script 13

After Execution - Post-Trade Tasks

Automated:

- Run Script 13 (Execution Logger)
 - Record all fills
 - Update portfolio state
 - Set initial stops for new positions

Quarterly Workflow

End of Quarter - Comprehensive Review

Performance Analysis (1-2 hours)

Manual:

- Run Script 22 (Performance Attribution)
 - Analyze return sources (asset class, sector, position)
 - Calculate alpha vs beta
 - Identify best/worst performers

Run Script 23 (Risk Analytics)

- Review VaR and CVaR
- Check tracking error and information ratio
- Analyze correlation matrix
- Evaluate factor exposures

Run Script 24 (Performance Dashboard)

- Generate quarterly report
- Compare vs benchmark
- Review rolling metrics

Strategy Review (1-2 hours)

Manual:

- Evaluate strategy performance
 - Are we meeting return targets? (Target: 10-15% annual)
 - Is risk within acceptable range? (Target: Max DD < 20%)
 - Are we generating alpha? (Target: >3% vs benchmark)

Review operational efficiency

- Are circuit breakers functioning?
- Any data quality issues?
- Any execution issues (slippage, unfilled orders)?

Document findings

- What worked well
- What needs improvement
- Any parameter adjustments needed

Parameter Review (optional)

If performance below targets for 2+ consecutive quarters:

- Run Script 17 (Walk-Forward Optimizer)
 - Test current parameters on recent data
 - Evaluate alternative parameter sets
 - Check stability ratios

If better parameters found and stable:

- Document proposed changes
- Run Monte Carlo simulation (Script 18)
- If outcomes acceptable:
 - Implement new parameters
 - Document change in strategy log

Annual Workflow

Year-End - Comprehensive Backtesting & Audit

Full System Backtest (2-3 hours)

Automated:

- Run Script 16 (Backtest Engine)
 - Backtest last 5 years of data
 - Compare to benchmark (buy-and-hold SPY)
 - Generate performance metrics
 - Identify regime-specific performance

Manual:

- Analyze backtest results
 - Are live results matching backtest expectations?
 - Any significant deviations?
 - Were there regime changes we didn't handle well?

Walk-Forward Optimization (3-4 hours)

Automated:

- Run Script 17 (Walk-Forward Optimizer)

- Optimize on 5-year rolling windows
- Test multiple parameter sets
- Calculate stability ratios

Manual:

- Review optimization results
- Are current parameters still optimal?
- Do alternative parameters show better stability?
- What is the degradation from in-sample to out-of-sample?

Monte Carlo Analysis (1-2 hours)

Automated:

- Run Script 18 (Monte Carlo Simulator)
- 10,000 simulations of trade randomization
- Calculate outcome distribution
- Determine confidence intervals

Manual:

- Analyze simulation results
- What is the range of expected outcomes?
- What is the probability of drawdown > 25%?
- Are we comfortable with the worst-case scenarios?

Tax Preparation

Manual:

- Run Script 13 report (all executions for year)
- Export to CSV
- Calculate realized gains/losses
- Separate short-term vs long-term
- Provide to tax accountant

Annual Report

Manual:

- Compile annual performance report
- Full-year return vs benchmark
- Risk-adjusted metrics (Sharpe, Sortino, Calmar)
- Drawdown analysis
- Attribution analysis
- Top 10 winners/losers
- Trade statistics (count, win rate, profit factor)
- Lessons learned
- Plan for next year

IMPLEMENTATION ROADMAP

Phase 1: Data Foundation (Weeks 1-2)

Week 1: Data Acquisition

- Implement Script 1 (EODHD Bulk Downloader)
 - Set up EODHD API account
 - Create exchanges.json config
 - Test INITIAL mode (full 400-day download)
 - Test INCREMENTAL mode (delta updates)
 - Verify new symbol discovery and delisting detection
- Implement Script 2 (Yahoo Finance Fundamentals)
 - Set up yfinance library
 - Test symbol mapping (EODHD → Yahoo format)
 - Download fundamentals for all symbols

Week 2: Data Processing

- Implement Script 3 (Data Consolidator & Validator)
 - Merge EODHD + Yahoo data
 - Implement all validation rules
 - Test with known bad data (edge cases)
 - Generate data quality report
- Implement Script 4 (Universe Screener)
 - Create filter_thresholds.json config
 - Apply all filters (price, volume, market cap, history)

- Test with multiple dates
- Verify qualified symbol counts
- Implement Script 5 (Indicator Calculator)
 - Calculate SMA_50, SMA_200, ATR_20, ADX_14
 - Verify calculations against known values
 - Test edge cases (insufficient data, gaps)

Deliverable: Data pipeline producing validated, indicator-enriched time series

Phase 2: Strategy Logic (Weeks 3-4)

Week 3: Signal Generation

- Implement Script 6 (Trend Qualifier)
 - Apply 3-part qualification (golden cross, price > SMA_50, ADX > 20)
 - Test with various market conditions
 - Verify trend-qualified counts
- Implement Script 7 (Momentum Ranker)
 - Calculate momentum scores
 - Sort descending
 - Test ranking stability over time
 - Verify top N selection

Week 4: Risk Management

- Implement Script 8 (Position Sizer)
 - Implement volatility-adjusted sizing
 - Apply floor (0.5%) and ceiling (8.0%)
 - Test with various account sizes
 - Verify share quantity calculations
- Implement Script 9 (Stop-Loss Calculator)
 - Calculate initial stops (3× ATR)
 - Calculate trailing stops (4× ATR)
 - Test activation logic (+15% profit threshold)
 - Verify weekly update logic (Friday only)
- Implement Script 10 (Exit Signal Generator)
 - Implement all 4 exit conditions
 - Test prioritization logic
 - Verify stop-loss checks
 - Test trend reversal and weakness detection

Deliverable: Complete signal generation and risk management system

Phase 3: Portfolio Management (Week 5)

Week 5: Rebalancing & Reporting

- Implement Script 11 (Monthly Rebalancer)
 - Integrate all previous scripts
 - Implement circuit breaker checks
 - Generate rebalancing recommendations
 - Test with multiple rebalancing dates
 - Verify exit/entry logic
- Implement Script 12 (Recommendation Report Generator)
 - Create PDF report template
 - Implement all report sections
 - Add charts (allocation, risk metrics)
 - Test report generation
 - Verify human-readable format
- Test end-to-end rebalancing workflow
 - Run full pipeline from data download to report
 - Verify all components integrate correctly
 - Check execution timing (fits within market hours)

Deliverable: Monthly rebalancing system with human-in-the-loop approval

Phase 4: Execution & Monitoring (Week 6)

Week 6: Operations

- Implement Script 13 (Execution Logger)
 - Design portfolio_state.json schema
 - Implement trade logging
 - Update portfolio state after fills
 - Test with simulated trades
- Implement Script 14 (Daily Risk Monitor)
 - Implement all 6 monitoring checks
 - Set up alert system (email/SMS)
 - Test alert triggers
 - Verify daily execution timing
- Implement Script 15 (Technical Chart Generator)
 - Create interactive HTML dashboard
 - Test with multiple symbols
 - Verify 200-day lookback
 - Test sidebar navigation
- Set up automation
 - Create cron jobs (daily data update, daily monitoring)
 - Test automated execution
 - Verify logging and error handling

Deliverable: Complete operational system with execution logging and monitoring

Phase 5: Backtesting Framework (Weeks 7-8)

Week 7: Backtest Engine

- Implement Script 16 (Backtest Engine)
 - Walk-forward simulation logic
 - Transaction cost and slippage modeling
 - Corporate action handling
 - Performance metrics calculation
 - Test on historical data (2019-2024)
 - Verify no look-ahead bias
- Run initial backtests
 - 5-year backtest (2019-2024)
 - Calculate all performance metrics
 - Compare to benchmark (SPY buy-and-hold)
 - Generate backtest report
 - Analyze results for issues

Week 8: Optimization & Monte Carlo

- Implement Script 17 (Walk-Forward Optimizer)
 - Parameter grid definition
 - In-sample optimization logic
 - Out-of-sample validation logic
 - Stability ratio calculation
 - Test with multiple windows
- Implement Script 18 (Monte Carlo Simulator)
 - Trade sequence randomization
 - 10,000 simulation execution
 - Confidence interval calculation
 - Outcome distribution analysis
- Run full optimization and simulation
 - Optimize all parameters
 - Validate stability
 - Run Monte Carlo simulations
 - Generate risk distribution report

Deliverable: Validated strategy with optimized parameters and risk assessment

Phase 6: Performance Management (Weeks 9-10)

Week 9: Attribution & Risk Analytics

- Implement Script 22 (Performance Attribution)
 - Asset class attribution
 - Sector attribution
 - Position-level attribution
 - Alpha vs beta decomposition
 - Test with historical data
- Implement Script 23 (Risk Analytics)
 - VaR and CVaR calculation
 - Tracking error and information ratio
 - Concentration metrics (Herfindahl index)
 - Correlation analysis
 - Factor exposure analysis
 - Test with various portfolios

Week 10: Dashboard & Integration

- Implement Script 24 (Performance Dashboard)
 - Equity curve chart
 - Monthly return heatmap
 - Drawdown chart
 - Rolling Sharpe ratio chart
 - Asset allocation pie chart
 - Position P&L table
 - Interactive features
- Final integration testing
 - Run complete end-to-end workflow
 - Test all scripts in sequence
 - Verify data flows correctly
 - Check performance (execution time)
 - Fix any integration issues

Deliverable: Complete performance management system with interactive dashboard

Phase 7: Production Deployment (Weeks 11-12)

Week 11: Production Setup

- Server/infrastructure setup
 - Choose hosting (cloud VM, dedicated server)
 - Install all dependencies
 - Set up Python environment
 - Configure database (if needed)
 - Set up automated backups
- Security setup
 - API key management (.env file)
 - File permissions
 - Network security
 - Backup encryption
- Documentation
 - Operator manual (how to run each script)
 - Troubleshooting guide
 - Emergency procedures
 - Contact information

Week 12: Testing & Launch

- Paper trading period (1 month)
 - Run system with real data
 - Generate recommendations (but don't execute)
 - Track how recommendations would perform
 - Verify all automations work
 - Fix any issues found
- Final review
 - Check all scripts operational
 - Verify data pipeline stable
 - Test alert system

- Review operator manual
- Sign-off from all stakeholders
- Go-live
 - Execute first real rebalancing
 - Monitor closely for first month
 - Document any issues
 - Make adjustments as needed

Deliverable: Live production system managing real capital

Success Criteria

Data Layer:

- ✓ Data updates automatically every trading day
- ✓ New symbols discovered within 1 day of listing
- ✓ Delisted symbols flagged within 1 day
- ✓ Data validation catches >99% of quality issues
- ✓ No gaps in historical data

Strategy Layer:

- ✓ Signals generated correctly (match manual calculation)
- ✓ Position sizing respects floor/ceiling limits
- ✓ Stop-losses calculated accurately
- ✓ Exit signals trigger at correct times

Portfolio Layer:

- ✓ Monthly rebalancing completes within 4 hours
- ✓ Reports are human-readable and actionable
- ✓ All trades logged accurately
- ✓ Portfolio state always current

Backtesting:

- ✓ Backtest matches forward test (no look-ahead bias)
- ✓ Transaction costs realistic
- ✓ Parameters validated through walk-forward
- ✓ Monte Carlo shows acceptable risk range

Validation & Decision:

- ✓ All 10 primary validation tests pass
- ✓ Overall score ≥ 19 points (ACCEPTABLE or better)
- ✓ No critical red flags present
- ✓ Out-of-sample stability ratio > 0.7
- ✓ Beats benchmark on risk-adjusted basis
- ✓ All required approvals obtained (PM, Risk, Compliance)

Performance Management:

- ✓ Attribution analysis identifies return sources
- ✓ Risk metrics updated daily
- ✓ Dashboard loads in <5 seconds
- ✓ All charts render correctly

Operations:

- ✓ System runs unattended 95%+ of time
- ✓ Alerts trigger appropriately (not too many, not too few)
- ✓ Recovery from errors automatic
- ✓ Execution slippage $< 0.1\%$ on average

APPENDIX: KEY FORMULAS SUMMARY

Technical Indicators

1. **SMA:** $SMA(n) = \sum Close[i] / n$
2. **ATR:** $ATR(n) = SMA(TR, n)$ where $TR = \max(H-L, |H-C_{prev}|, |L-C_{prev}|)$
3. **ADX:** $ADX = EMA(DX, n)$ where $DX = 100 \times |+DI - -DI| / (+DI + -DI)$

Strategy Rules

- 4. **Trend Qualification:** $SMA_{50} > SMA_{200}$ AND $Close > SMA_{50}$ AND $ADX > 20$
- 5. **Momentum Score:** $(Close - SMA_{200}) / SMA_{200} \times 100$
- 6. **Position Size:** $\text{floor}(\text{Base_Risk} \times \text{Vol_Multiplier} / \text{Price})$ where $\text{Base_Risk} = \text{Equity} \times 2\%$
- 7. **Initial Stop:** $\text{Entry_Price} - (3.0 \times \text{ATR})$
- 8. **Trailing Stop:** $\text{max}(\text{Current_Stop}, \text{Price} - (4.0 \times \text{ATR}))$ if $\text{Profit} > 15\%$

Risk Management

- 9. **Max Position:** $\text{Equity} \times 8.0\%$
- 10. **Min Position:** $\text{Equity} \times 0.5\%$
- 11. **Drawdown:** $(\text{Equity} - \text{Peak_Equity}) / \text{Peak_Equity}$
- 12. **Circuit Breaker:** Halt if $\text{Drawdown} < -15\%$

Performance Metrics

- 13. **Sharpe Ratio:** $(\text{Return} - \text{RFR}) / \text{Volatility}$
- 14. **Max Drawdown:** $\min(\text{Equity} / \text{Peak_Equity} - 1)$
- 15. **Win Rate:** $\text{Winning_Trades} / \text{Total_Trades}$
- 16. **Profit Factor:** $\text{Gross_Profit} / \text{Gross_Loss}$

Attribution

- 17. **Asset Class Contribution:** $\text{Weight} \times \text{Return}$
- 18. **Alpha:** $\text{Return} - (\text{Beta} \times \text{Benchmark_Return})$
- 19. **Information Ratio:** $\text{Active_Return} / \text{Tracking_Error}$
- 20. **VaR:** $\text{Equity} \times \text{Percentile}(\text{Returns}, 95\%)$

END OF DOCUMENT

Version: 3.2 Executive Summary (Updated with Validation Framework)
Date: February 2026
Purpose: Complete system architecture without code - from design to deployment decision
Status: Production-Ready Blueprint with Validation Methodology

Total System Components:

- 24 Scripts (Data: 5, Strategy: 5, Portfolio: 2, Operations: 3, Backtest: 3, Validation: 3, Performance: 3)
- 20 Key Formulas (all explicitly defined)
- 10 Primary Validation Tests (Go/No-Go criteria)
- 5-Tier Decision Matrix (deployment framework)
- 4-Phase Operational Workflow (daily/weekly/monthly/quarterly)
- 7-Phase Implementation Roadmap (12 weeks)

Validation Framework Included:

- 10 mandatory validation tests (all must pass)
- 30-point scoring system (quantitative assessment)
- 7 critical red flags (instant disqualification criteria)
- Out-of-sample validation requirements (stability ratio >0.7)
- Monte Carlo validation thresholds (tail risk assessment)
- 5-tier deployment decision matrix
- 10-step recommendation process
- Complete documentation requirements

Next Steps:

- 1. Review and approve architecture
- 2. Begin Phase 1 implementation (Data Foundation)
- 3. Follow 12-week roadmap to production
- 4. Run backtest suite (Scripts 16-18)
- 5. Apply validation framework systematically
- 6. Make Go/No-Go decision using decision matrix
- 7. Obtain required approvals (PM, Risk, Compliance)
- 8. Deploy according to tier assignment
- 9. Paper trade for 1 month before going live